Retention

Week 4, Lecture 1 — B2C to $10k MRR

B2C to $10k MRR: A 10-Week Growth Manual · Autumn 2026

What we will cover

- Why D1, D7, and D30 each measure something different
- The three retention curve shapes and what each predicts about your MRR
- Net revenue retention vs logo retention
- The Hook Model: what survives scrutiny, what does not

D1 / D7 / D30: three different questions

Day 1 retention measures onboarding, not the product

- **D1 = did the user come back the day after signing up?**
- High D1 (> 40%) means your activation flow delivered value fast
- Low D1 means users hit friction before seeing why they should return
- D1 is the metric most sensitive to onboarding changes
- Fixing D1 is often faster and cheaper than fixing D30

Day 7 retention measures habit formation

- **D7 = did the user return in the first week?**
- Seven days is one full natural cycle: weekly meeting, weekly workout, weekly budget
- If your product is meant to be a weekly habit, D7 is your primary signal
- D7 < 20% on a product with a weekly use-case is a warning
- Lenny Rachitsky (2020): consumer SaaS targets 40% good, 70% great at 6 months

Day 30 retention measures product-market fit

- **D30 = did the user survive the first month?**
- RevenueCat data (2025): monthly subscription year-1 retention is 17% median
- A flattening D30 curve is one of the strongest PMF signals available
- PostHog (Vandervell, 2023): retention curves that flatten signal product-market fit
- D30 is too slow to guide week-to-week decisions; use D1/D7 for that

Retention benchmarks by category (Rachitsky 2020)

Good

- Consumer social: 25% at 6 months
- Consumer transactional: 30% at 6 months
- Consumer SaaS: 40% at 6 months

Great

- Consumer social: 45% at 6 months
- Consumer transactional: 50% at 6 months
- Consumer SaaS: 70% at 6 months

The retention curve

Three curve shapes, three predictions

- **Smiling:** steep early drop, then flattens. A core of habitual users survives.
- **Flat:** steady gentle decline. Consistent usage, no habit lock-in.
- **Decaying:** continuous steep drop. Most users never come back.

The shape predicts your MRR trajectory better than any single data point.

A decaying curve + paid acquisition = burning money. Fix retention first.

Smiling vs decaying: two real-data patterns

Smiling (PMF signal)

```
D1   D7   D30
55%  28%  22%  <- flattens
54%  29%  23%
```

Gap between D7 and D30 is small.

Decaying (acquisition trap)

```
D1   D7   D30
60%  30%   8%  <- still falling
59%  25%   5%
```

D30 still dropping. Scale = accelerate churn.

Net revenue retention

Logo retention vs net revenue retention

- **Logo retention:** what share of customers are still paying after 30 days?
- **Net revenue retention (NRR):** what share of the revenue from those customers are you still collecting, including expansions, contractions, and churn?
- NRR > 100% = expansion revenue exceeds churn. MRR grows even if you acquire zero new customers.
- Temperton (2024): gross vs net revenue churn distinction; net negative churn is the goal

NRR formula

```
NRR = (Start MRR + Expansion − Contraction − Churn) / Start MRR x 100
```

Example: start $8,000, +$1,200 expansion, -$400 contraction, -$600 churn

```
NRR = (8,000 + 1,200 − 400 − 600) / 8,000 x 100 = 102.5%
```

NRR above 100% means the business compounds even without new acquisitions.

Habit loops

The Hook Model (Eyal, 2014): four steps

1. **Trigger** (external: notification, email; internal: boredom, anxiety)
2. **Action** (minimum-effort behavior: open app, tap, scroll)
3. **Variable reward** (unpredictable payoff: new content, social validation, progress)
4. **Investment** (user puts something in: data, settings, relationships, reputation)

Investment increases the probability the next trigger fires.
That is how the loop compounds.

What survives scrutiny from the Hook Model

- Variable rewards are real: intermittent reinforcement produces stronger habits than fixed rewards (Skinner schedule research, well-replicated)
- Investment effects are real: sunk cost + data lock-in both reduce churn
- External triggers (push notifications) work at D1-D7; they decay after day 30 if no internal trigger forms
- Internal triggers (anxiety, boredom, habit) are what sustain D30+ retention

What the Hook Model does not solve

- If your product does not deliver genuine value, the loop does not close
- Variable rewards without real utility create engagement without retention
- Manufactured urgency (dark patterns) produces short-term D1 spikes and long-term D30 collapses
- PostHog (Vandervell, 2023): high engagement without retention signals wrong metrics, not PMF

> "Retention is widely considered to be the most important metric, yet it remains poorly understood."

<footer>Lenny Rachitsky, "What is good retention?", 2020</footer>

Take-aways

1. D1 measures onboarding; D7 measures weekly habit; D30 measures product-market fit.
2. Name your retention curve shape before making any acquisition investment.
3. NRR above 100% means your existing customers compound your MRR without new signups.
4. The Hook Model is a useful diagnostic: find where the loop breaks to find why D7 drops.
5. Benchmarks (Rachitsky 2020): consumer SaaS 40% good / 70% great at 6 months.

What is next

- **Section (this week):** build the cohort table in SQL or PostHog; name your curve shape
- **Lecture 2 (this week):** cohort analysis in depth; power-user query; churn diagnosis
- **Week 4 reading:** retention, cohort analysis, and habit loops
- **HW 2 due:** pricing and paywall (check the syllabus for submission details)